TECHNICAL MANUAL / DOCKER PACKAGE 4.1.2

T-Delayer

Package generation for Ubuntu 24.04 LTS and 26.04 LTS on amd64.

Build packages

From the repository root on a machine with Python 3 and a working Docker CLI/daemon:

```
./packaging/build-deb.sh --ubuntu all --version 4.1.2
# Or build one release:
./packaging/build-deb.sh --ubuntu 24.04 --version 4.1.2
./packaging/build-deb.sh --ubuntu 26.04 --version 4.1.2
```

The build machine needs network access to Ubuntu registries/APT repositories, several GB of free space, and support for `linux/amd64` containers. On ARM build machines, Docker must provide amd64 emulation (Docker Desktop normally does). All compilation and `dpkg-deb` operations run in Linux containers. The C debug and foreground release builds and `read_chunk_file` build run as part of the image build. The final image includes the release binary and English web UI.

Output defaults to `build-artifacts/deb/`; change it with `--output DIRECTORY`:

```
t-delayer-docker_4.1.2-1ubuntu24.04_amd64.deb
t-delayer-docker_4.1.2-1ubuntu26.04_amd64.deb
```

Each package contains a complete single-container image matching its Ubuntu release, the image content manifest, installer questions, service, helper, and the installation manual. Installation loads that bundled image; it does not compile source code or pull an application image from a registry. APT still needs repositories for missing host dependencies unless you have provisioned them beforehand.

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Installer questions and web-bind

The English debconf template `t-delayer-docker/web-bind` is a string with default `127.0.0.1`, labelled **Web UI bind address (web-bind):**. The package config script asks it in its own step after the Docker network questions. Its priority is `critical`, so an attended frontend displays it even when the frontend is set to show only critical questions. Other network and runtime questions use `high`.

Debconf's normal seen flags are retained. APT can run the config script during preconfiguration and again during package configuration; resetting these flags would repeat the same question. Previously configured installations retain their answers on upgrade. To show all questions again, use:

```
sudo dpkg-reconfigure -f readline -p low t-delayer-docker
```

The noninteractive frontend continues to use preseeded or default values without prompting. The post-install script reads the debconf answer into `web_bind` in `/etc/t-delayer/config.json`. Reconfiguration validates the result and recreates the container while preserving persistent data and credentials.

In host mode the address controls nginx's listener. In bridge mode it controls Docker's published host address; nginx listens on port 8080 inside the container. Macvlan/ipvlan require `0.0.0.0` and expose the chosen port at the container IP. The bind address does not select the multicast interface.

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Installer verification

Run the unit tests and disposable package tests described in `packaging/tests/README.md`. The attended test drives a real terminal with the readline frontend, real dpkg/debconf and the package maintainer scripts. It checks fresh installation, loopback/all-interface/custom web-bind values, critical priority, host/bridge listener generation, repeated config execution, unattended preseeding, and data retention. Docker and systemd are test doubles in this test.

Both unchanged 4.1.2 amd64 packages passed fresh attended installation in full Ubuntu 24.04 and 26.04 VMs on 12 November 2026, using real Docker and systemd. All five installer prompts appeared exactly once and their saved answers were verified. Both guests then passed a full reboot with the recording mount held: Docker ran while the container remained stopped and the application start job waited. Releasing the mount triggered automatic healthy startup, with database, recording, credential and authentication data preserved. The complete evidence and earlier 4.1.1 results are recorded in `packaging/tests/VM-VALIDATION.md`.